

EXPERIMENT 11

Deploying a Zoho Creator Application from Development to Production Environment

Environment-Based Promotion for the Patient Appointment Scheduling System

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Aim

To demonstrate the process of promoting a Zoho Creator application from its Development environment to Production, using the Patient Appointment Scheduling System, with emphasis on environment isolation, controlled publishing, and protecting live data from untested changes.

Introduction

Deploying an application safely is as important as building it. In a low-code platform such as Zoho Creator, a form, workflow or report can be edited at any time, so without a controlled deployment process a change made during routine development could reach real users immediately, before it has been reviewed or tested.

Zoho Creator addresses this through built-in Environments. Development, Stage and Production are kept as separate, isolated workspaces, each capable of holding its own copy of the application and its own data. Work is built and tested in Development; only once it is reviewed is it explicitly published to Production, where real end users interact with it. This separation keeps experimentation safe and keeps the live application stable.

Application Used

The practical was performed in Zoho Creator using the Patient Appointment Scheduling System application. The application's built-in Environments feature (Development, Stage, and Production) was used to move the application from the Development workspace, where it is actively built and modified with sample records, towards a Production workspace intended for real end users. No upgrade-only feature is claimed as enabled or configured in this record; deployment is demonstrated at the level accessible within a trial/developer account.

Deployment Concept

The key concept demonstrated is environment-based, staged deployment rather than editing an already-live application directly. Zoho Creator's Environments module keeps Development, Stage and Production as separate workspaces so changes can be built and tested without affecting the version real users rely on. Promotion between environments follows this path:

Development (build & test) → Review in Stage → Publish → Production (live)

- The developer builds and edits forms, workflows and reports inside the Development workspace.
- Before promotion, the application is reviewed in Development or Stage for correctness, using sample data that is separate from live records.
- The Publish action copies the reviewed application build into the Production environment, where it becomes available to end users.
- Access to each environment can be restricted by role, so only authorised users (for example, a Medical Administrator or CEO role) can publish to Production.
- This avoids exposing untested changes directly to live patient and appointment data.

Procedure

1. Open the Patient Appointment Scheduling System application in Zoho Creator in Edit mode.
2. Navigate to the DEPLOY section in the left panel and open Environments.
3. Observe the Environments page, which lists Stage and Production as separate environment columns alongside the application's current Development workspace (Figure 1).
4. Review the application currently in Development, where forms, reports and workflows have already been built and tested with sample records from earlier experiments.
5. Select Manage on the target environment to review environment-specific settings, such as data sources and integration endpoints, before publishing.
6. Select Publish to promote the reviewed Development build into the Production environment.
7. Confirm that Production now serves the application to end users, isolated from further Development changes until the next publish.

Environment	Purpose
Development	Active build-and-test workspace; sample records only; where forms, workflows and reports were created in earlier experiments.
Stage	Optional pre-production checkpoint for reviewing a build under near-live conditions before it is published.
Production	Live environment used by real end users (patients, front-desk staff, doctors); updated only through an explicit Publish action.

Environments Page of the Patient Appointment Scheduling System

Deployment Configuration

The configuration used to promote the application from Development to Production is summarised below. No custom script is required for this step; the promotion is carried out through the Manage and Publish controls of the Environments page.

Parameter	Detail
Source Environment	Development
Target Environment	Production
Promotion Method	Manage (review settings) followed by Publish
Access Restriction	Role-based; publish access limited to authorised roles such as CEO and Medical Administrator
Data Isolation	Production holds its own live data; Development's sample/test records are not carried into Production

Deployment Approach Comparison

The staged, environment-based approach used here is compared with a general, risk-prone approach below.

Approach	Steps
Before (direct-editing approach)	Edit the application directly in Production → changes take effect immediately for live users → no isolated testing → risk of exposing untested or broken functionality to real patient data
Staged (used in this practical)	Build and test in Development → review in Stage if needed → Publish the reviewed build to Production → live users only ever see tested, reviewed changes

DIRECT-EDITING APPROACH • Production edited directly • No separate test workspace • Changes go live instantly • Environments isolated: 0	STAGED DEPLOYMENT APPROACH • Development used for all changes • Stage available for pre-production review • Production updated only via Publish • Environments isolated: 3 (Dev / Stage / Production)
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Figure 2: Environments Used by the Direct-Editing and Staged Deployment Approaches (Conceptual Diagram)

No benchmark was performed in this practical, so no measured improvement in deployment time is claimed. The comparison is based on the number of isolated environments involved and the resulting exposure of live data to untested changes.

Deployment Techniques

General practices that contribute to safe deployment in Zoho Creator applications are given below.

No.	Technique	Explanation
1	Environment Isolation	Keep Development, Stage and Production as separate workspaces with their own data.
2	Staged Testing Before Publish	Review and test a build in Development/Stage before promoting it.
3	Role-Based Publish Access	Restrict the Publish action to authorised roles only.
4	Environment-Specific Data Sources	Keep Production's live data separate from Development's test records.
5	Rollback / Version Awareness	Review a build carefully before it overwrites what is live in Production.

Status key: shaded green rows (Environment Isolation, Staged Testing, Environment-Specific Data Sources) were directly used in this practical; shaded amber rows (Role-Based Publish Access, Rollback/Version Awareness) are general practices consistent with the role-based access control configured in an earlier experiment, but were not re-verified as part of this specific deployment.

Figure 3: Deployment Techniques and Their Status (Conceptual Diagram)

Optimized Deployment Workflow

Figure 4 is a conceptual flowchart of the staged deployment workflow used to promote the application:

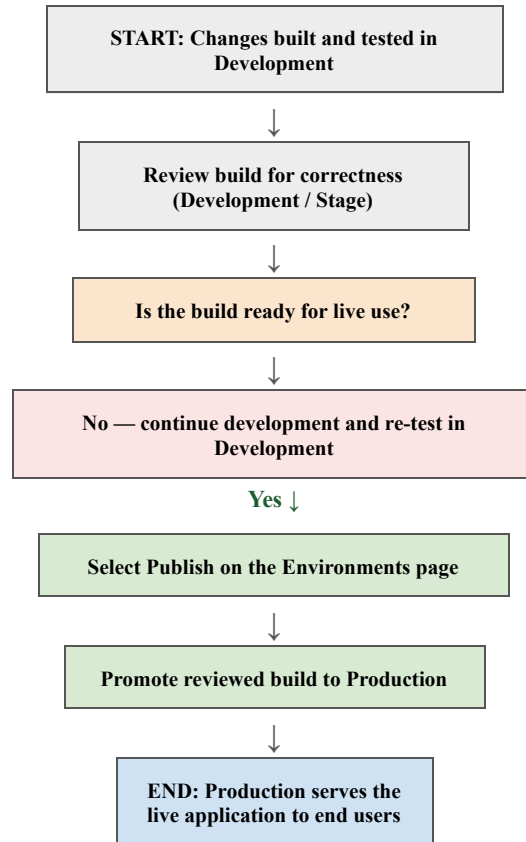


Figure 4: Staged Deployment Workflow for the Patient Appointment Scheduling System (Conceptual Diagram)

- Build and test: all form, report and workflow changes are made and verified in Development first.
- Readiness check: the build is reviewed to confirm it behaves correctly before it is allowed to reach Production.
- Publish: the explicit Publish action on the Environments page is the only way a change reaches Production.
- Live service: once published, Production serves the application to real patients and staff, isolated from any further Development changes until the next publish.

Conclusion

Safe deployment is important in low-code applications because reliability depends not only on how an application is built but also on how changes reach real users. In this practical, the Patient Appointment Scheduling System was promoted from Development to Production using Zoho Creator's built-in Environments feature, so that changes were reviewed before they affected live data.

Staged, environment-based deployment improves safety by avoiding direct edits to a live application and by keeping Development's test records separate from Production's real data. Practices such as environment isolation, staged testing before publish, role-based publish access, and environment-specific data sources help applications scale from a single developer's trial account to a system multiple real users depend on. These practices contribute to safe, auditable, and maintainable Zoho Creator deployments. No benchmark was performed, so the benefits are described in terms of reduced exposure of untested changes rather than measured deployment time.